



Desenvolvimento de Sistemas de Informação Analíticos

Business Process Analytics



2025 / 2026

Universidade do Minho

OBJECTIVE

- Analytical Information Systems?
 - Information System ... as a work system whose processes and activities are devoted to processing information, that is, capturing, transmitting, storing, retrieving, manipulating, and displaying information [Alter, 2008]
 - AIS are fundamental to address veracity and value when systematically supporting the data life cycle [Michalczyk and Scheu, 2020]
- Fundamental systems for supporting decision-making



Alter, S.: Defining information systems as work systems: implications for the IS field. European Journal of Information Systems 17(5), 448–469 (2008)

Michalczyk, S., Scheu, S.: Designing an analytical information system engineering method. In: Proceedings of the Twenty-Eighth European Conference on Information Systems (ECIS2020). Association for Information Systems (2020)

ANALYTICAL SYSTEMS

**Supporting Decision-making in
Organizations**



GENERAL INDICATORS

- Productivity (assess the use of available resources)
- Efficiency (produce the most with less resources)
- Efficacy (are processes working properly?)
- Quality (does production meet the quality standards?)
- Capacity (how much can be produced in a specific period?)
- Profitability (verify production profit compared to sales revenue)
- ...





Efficiency, profitability,
..., of business

order_number	customer	customer_register_date	Date	Time	price	product	product_weight	product_price	product_register_date	item	package	package_weight
o-990001	AlpenTech Innovations AG	03/04/2023 01:00	03/04/2023	12:08:00	1353.98	Echo	0.78	99.99	04/03/2023 01:00	i-880001	p-660001	8.484
o-990001	AlpenTech Innovations AG	03/04/2023 01:00	03/04/2023	12:08:00	1353.98	Echo	0.78	99.99	04/03/2023 01:00	i-880002	p-660001	8.484
o-990001	AlpenTech Innovations AG	03/04/2023 01:00	03/04/2023	12:08:00	1353.98	iPhone 11 Pro	0.188	1149.0	04/03/2023 01:00	i-880003	p-660001	8.484
o-990002	AlpenTech Innovations AG	03/04/2023 01:00	03/04/2023	16:36:00	3083.0	MacBook Pro	1.37	2500.0	04/03/2023 01:00	i-880004	p-660001	8.484
o-990002	AlpenTech Innovations AG	03/04/2023 01:00	03/04/2023	16:36:00	3083.0	Kindle Paperwhite	0.495	129.0	04/03/2023 01:00	i-880005	p-660001	8.484
o-990002	AlpenTech Innovations AG	03/04/2023 01:00	03/04/2023	16:36:00	3083.0	iPad mini	0.28	449.0	04/03/2023 01:00	i-880006	p-660001	8.484
o-990003	Celtica Green Farms Oy	03/04/2023 01:00	03/04/2023	23:31:00	3599.99	Fire Stick	0.2	39.99	04/03/2023 01:00	i-880007	p-660018	0.48
o-990003	Celtica Green Farms Oy	03/04/2023 01:00	03/04/2023	23:31:00	3599.99	iPhone 8	0.21	529.0	04/03/2023 01:00	i-880008	p-660003	3.606
o-990003	Celtica Green Farms Oy	03/04/2023 01:00	03/04/2023	23:31:00	3599.99	iPad Pro	0.483	1099.0	04/03/2023 01:00	i-880009	p-660003	3.606
o-990003	Celtica Green Farms Oy	03/04/2023 01:00	03/04/2023	23:31:00	3599.99	iPhone X	0.172	699.0	04/03/2023 01:00	i-880010	p-660003	3.606
o-990003	Celtica Green Farms Oy	03/04/2023 01:00	03/04/2023	23:31:00	3599.99	Kindle Paperwhite	0.495	129.0	04/03/2023 01:00	i-880011	p-660003	3.606
o-990003	Celtica Green Farms Oy	03/04/2023 01:00	03/04/2023	23:31:00	3599.99	iPad Pro	0.483	1099.0	04/03/2023 01:00	i-880012	p-660003	3.606
o-990004	Baltic Wave Energies Ltd.	03/04/2023 01:00	04/04/2023	12:30:00	2241.95	Echo	0.78	99.99	04/03/2023 01:00	i-880013	p-660002	3.125
o-990004	Baltic Wave Energies Ltd.	03/04/2023 01:00	04/04/2023	12:30:00	2241.95	Kindle Paperwhite	0.495	129.0	04/03/2023 01:00	i-880014	p-660002	3.125
o-990004	Baltic Wave Energies Ltd.	03/04/2023 01:00	04/04/2023	12:30:00	2241.95	Echo Show 8	0.98	129.99	04/03/2023 01:00	i-880015	p-660002	3.125
o-990004	Baltic Wave Energies Ltd.	03/04/2023 01:00	04/04/2023	12:30:00	2241.95	Echo Show 8	0.98	129.99	04/03/2023 01:00	i-880016	p-660014	1.463

Business Analytics

Process Analytics

Activity	Date	Time	Customer	Type	Category	Employee	Employee_Register_Date	Employee_Role	order_number
place_o-990001	03/04/2023	12:08:18	AlpenTech Innovations AG	place order	PlaceOrder				o-990001
pick_i-880003	03/04/2023	13:21:49		pick item	PickItem	Leah Tacke genannt Unterberg	03/04/2023 01:00	Warehousing	o-990001
pick_i-880001	04/04/2023	09:50:21		pick item	PickItem	Leah Tacke genannt Unterberg	03/04/2023 01:00	Warehousing	o-990001
pick_i-880002	04/04/2023	16:12:13		pick item	PickItem	Nina Graves	03/04/2023 01:00	Warehousing	o-990001
confirm_o-990001	04/04/2023	16:34:29		confirm order	ConfirmOrder	Wil van der Aalst	03/04/2023 01:00	Sales	o-990001
pay_o-990001	05/04/2023	17:47:04		pay order	PayOrder	Wil van der Aalst	03/04/2023 01:00	Sales	o-990001
create_p-660001	08/04/2023	01:00:00		create package	CreatePackage	Detlef Wetzeler	03/04/2023 01:00	Warehousing	o-990001
send_p-660001	10/04/2023	10:40:42		send package	SendPackage	Viki Peeva	03/04/2023 01:00	Shipment	o-990001
deliver_p-660001	11/04/2023	11:03:27		package delivered	PackageDelivered	Viki Peeva	03/04/2023 01:00	Shipment	o-990001
place_o-990002	03/04/2023	16:36:55		place order	PlaceOrder				o-990002
pick_i-880006	04/04/2023	10:01:31	AlpenTech Innovations AG	pick item	PickItem	Leah Tacke genannt Unterberg	03/04/2023 01:00	Warehousing	o-990002
confirm_o-990002	04/04/2023	16:09:41		confirm order	ConfirmOrder	Istvan Koren	03/04/2023 01:00	Sales	o-990002
pick_i-880004	05/04/2023	14:53:49		pick item	PickItem	Mahsa Bafrani	03/04/2023 01:00	Warehousing	o-990002
pick_i-880005	06/04/2023	16:31:52		pick item	PickItem	Christopher Schwanen	03/04/2023 01:00	Warehousing	o-990002
pay_o-990002	10/04/2023	13:00:09		pay order	PayOrder	Istvan Koren	03/04/2023 01:00	Sales	o-990002

Inventory

Objects

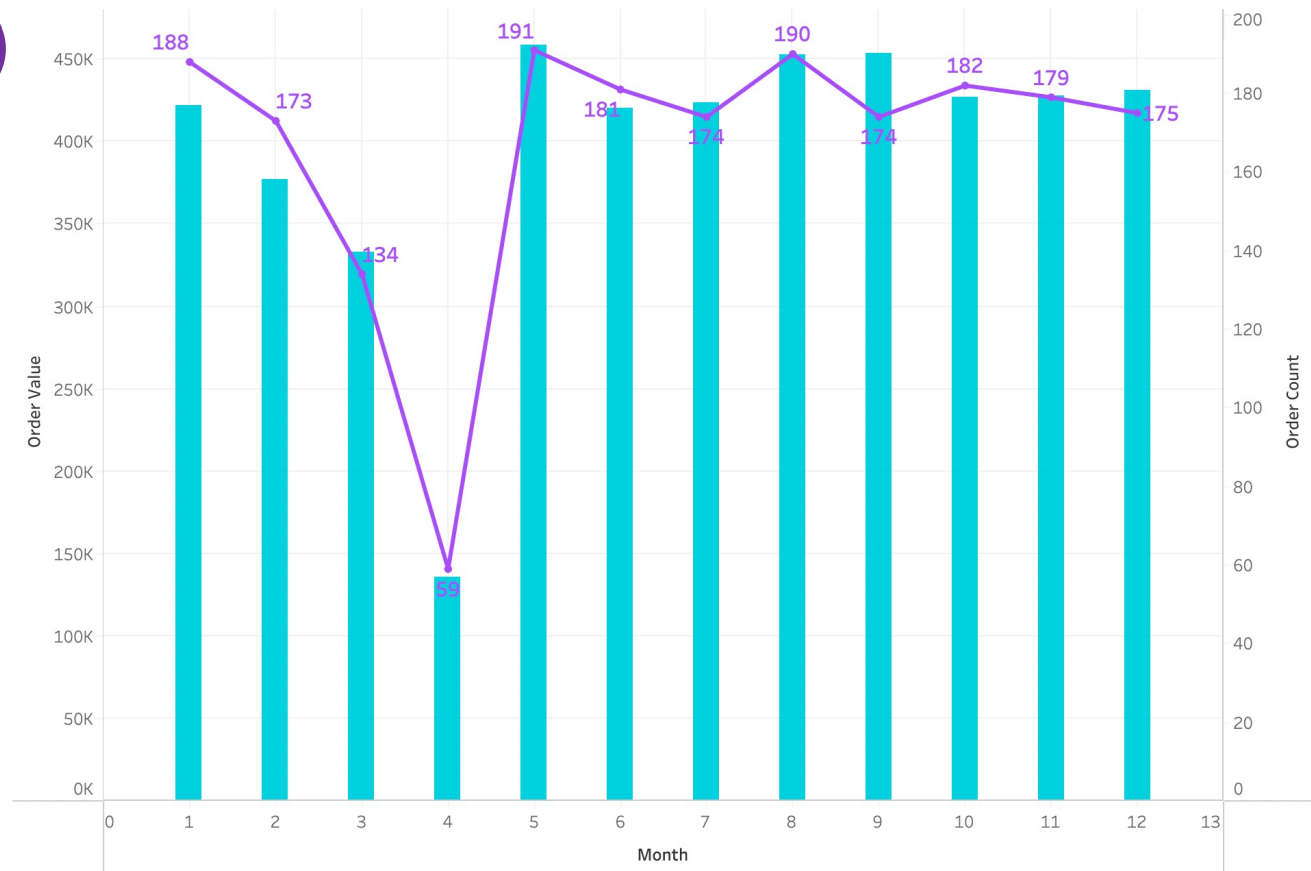
...

... OCEL 2.0 Standard
(Object-Centric Event Log)
<https://www.ocel-standard.org>

Business Analytics

Measure Names

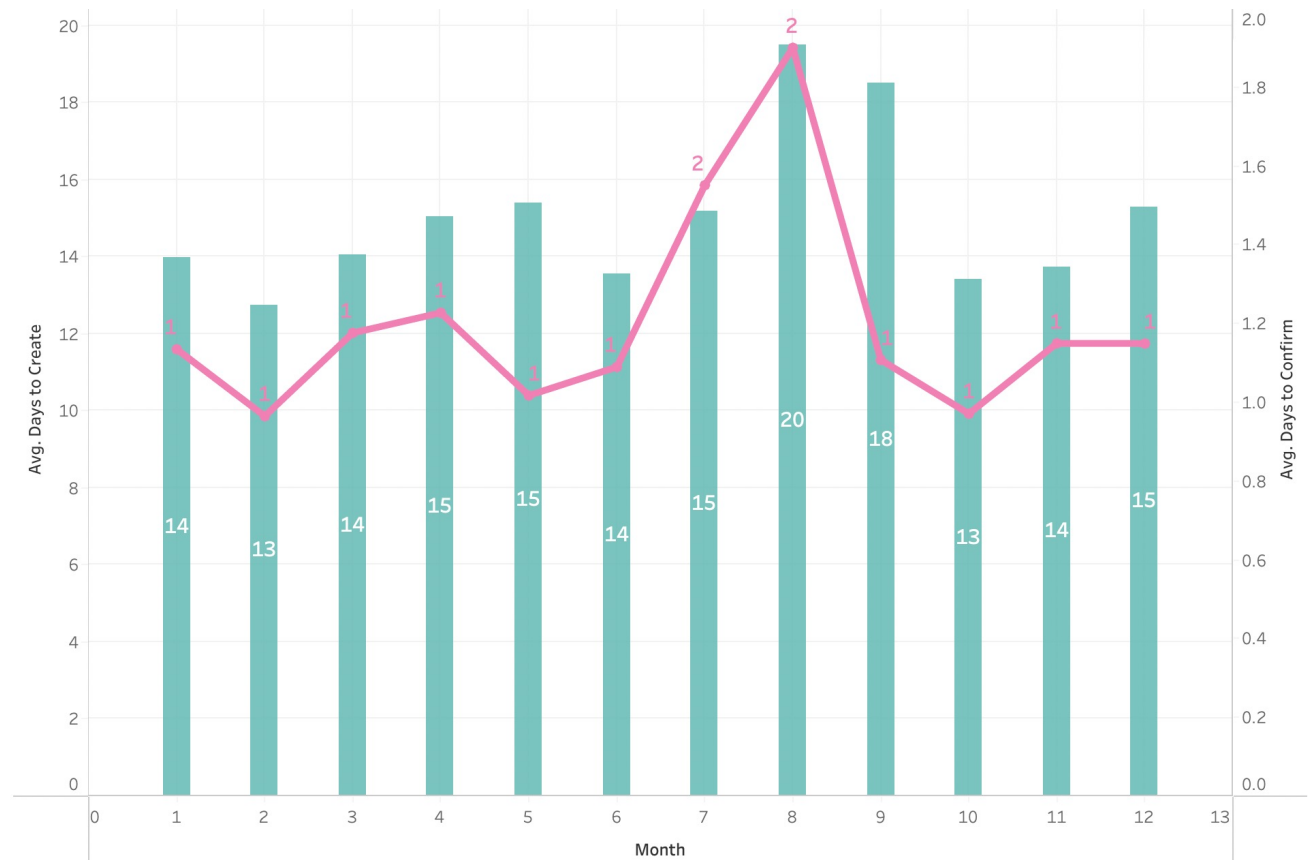
- Order Count
- Order Value



Process Analytics

Measure Names

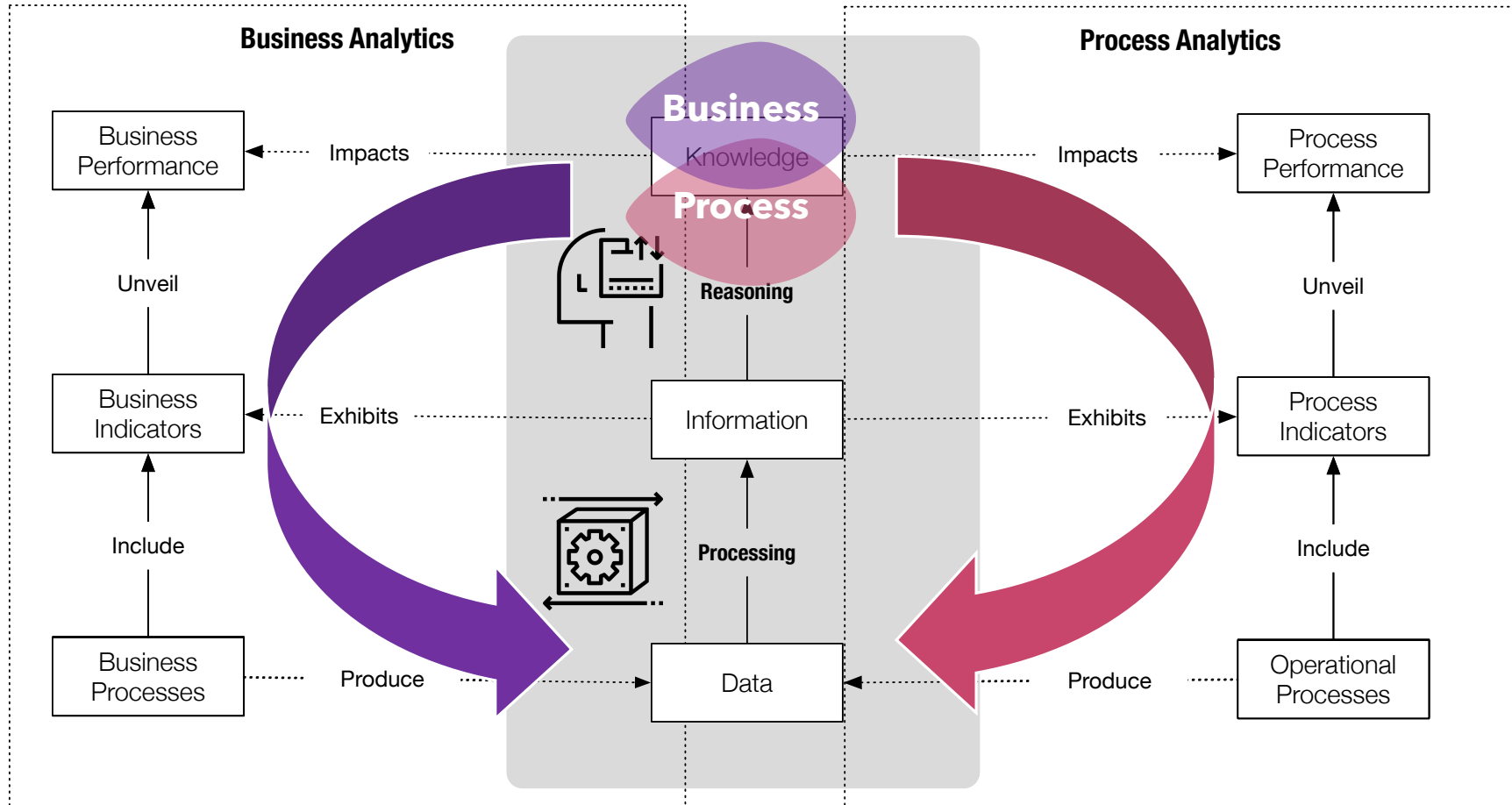
- Avg. Days to Confirm
- Avg. Days to Create

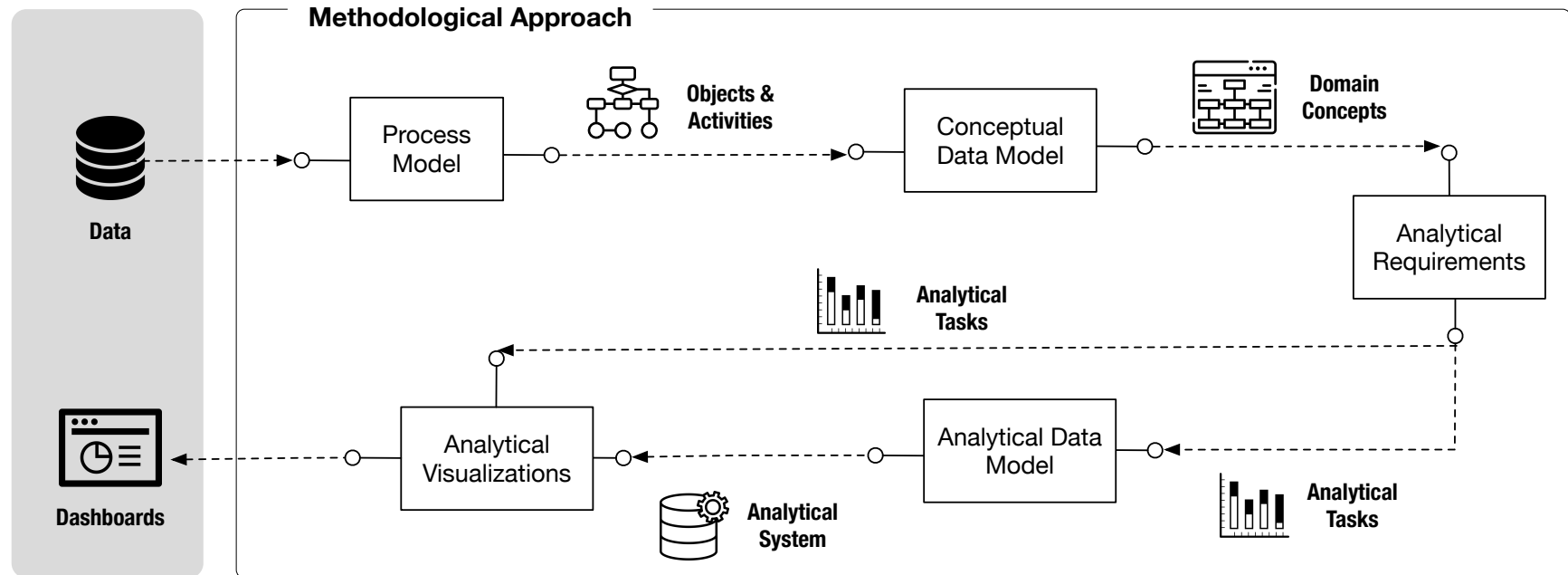


BUSINESS PROCESS ANALYTICS



Business Process Analytics



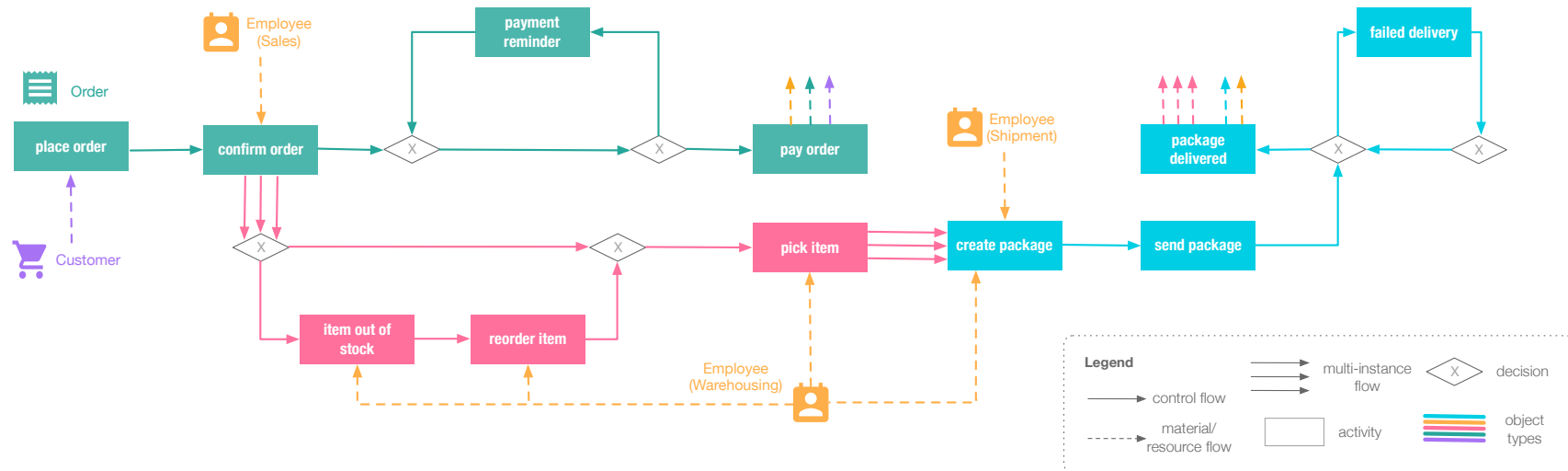
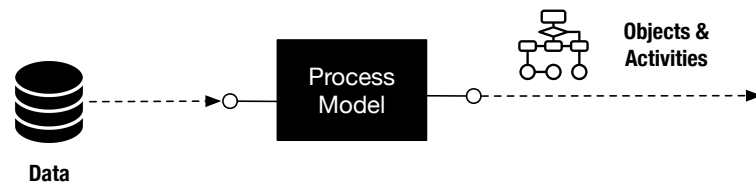


EXAMPLE

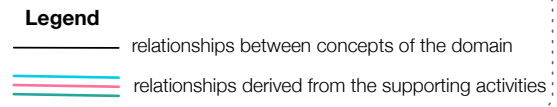
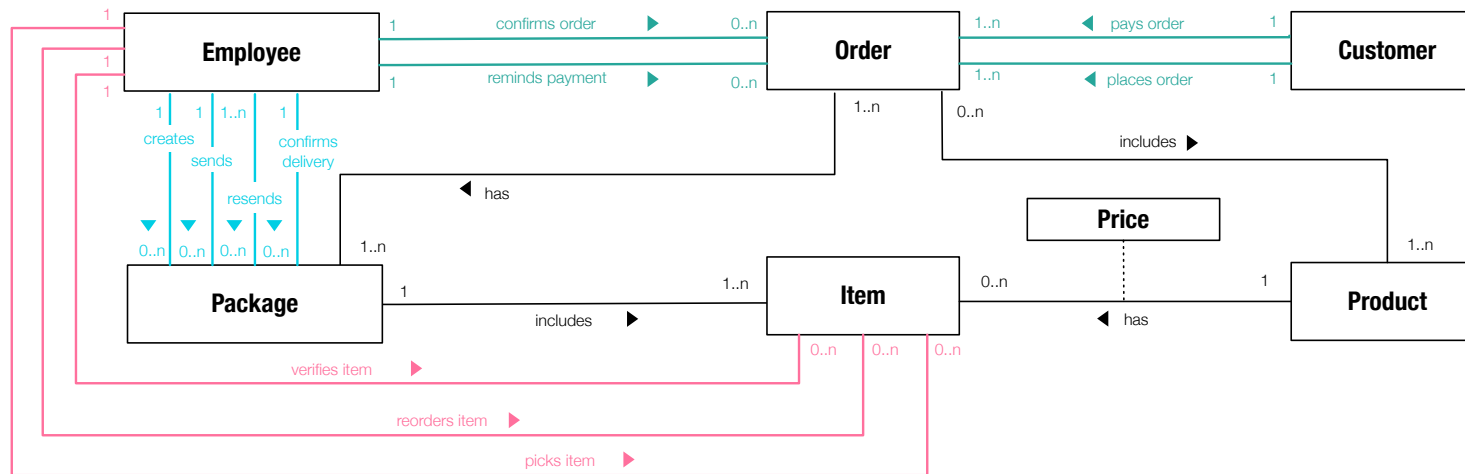


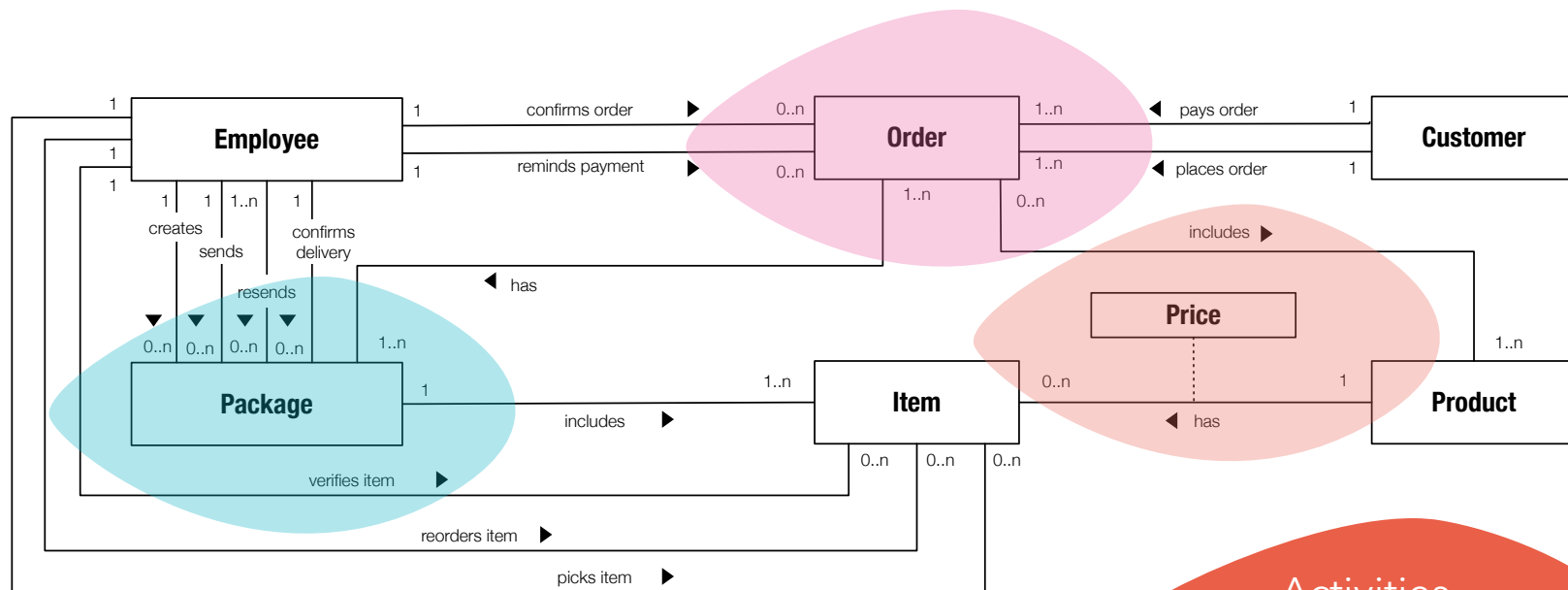
Management of
customers'
orders



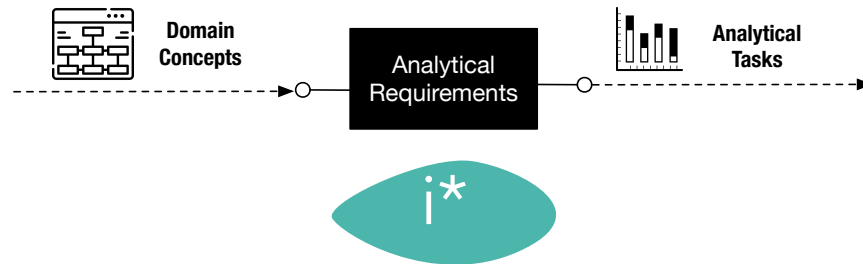


B. Knopp, W. M. van der Aalst (2023), Order management object-centric event log in OCEL 2.0 standard, [Data set], 2023.
doi:10.5281/zenodo.8428112.





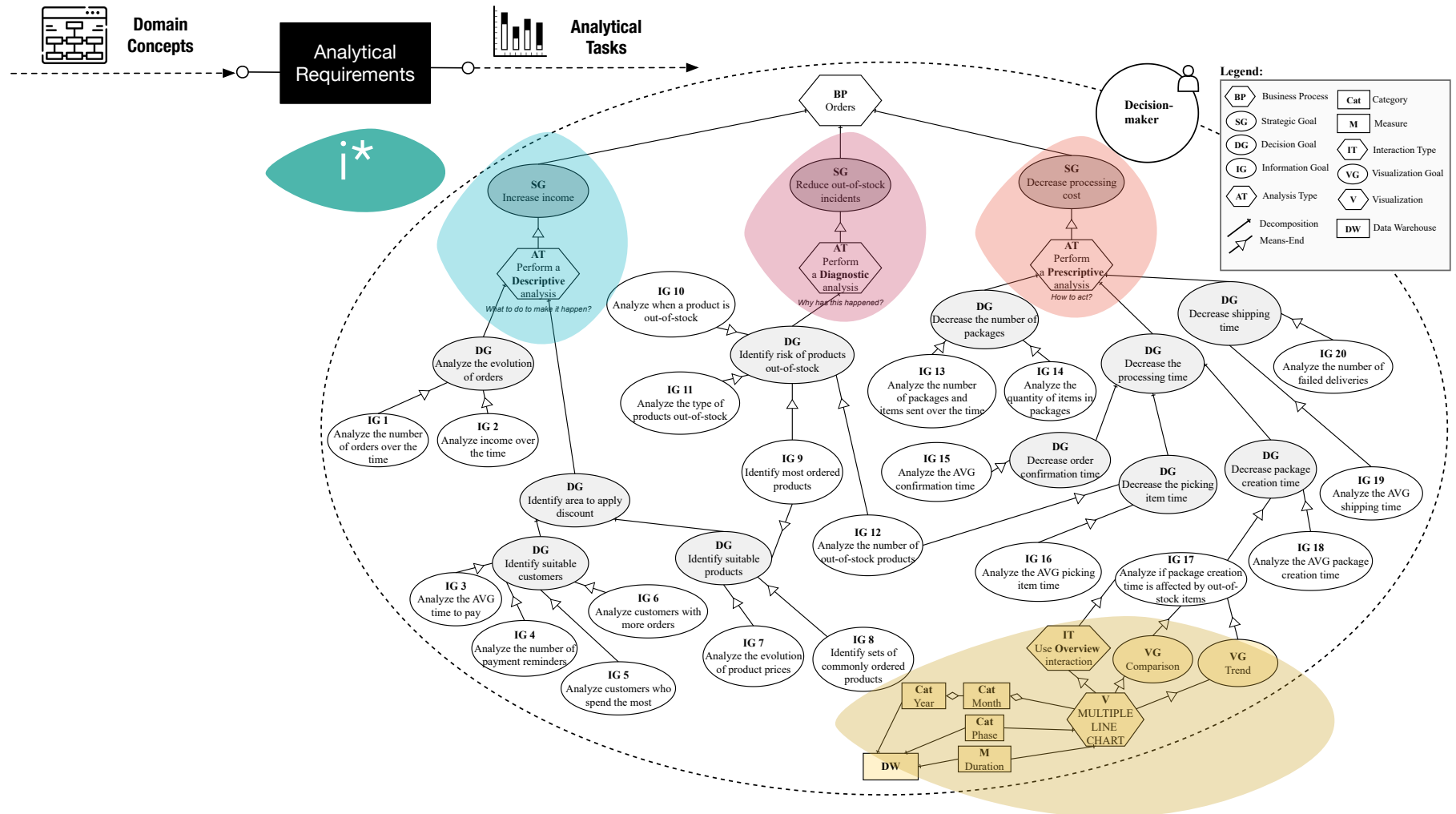
Activities
How long they took?
How many failures?
Can be improved?

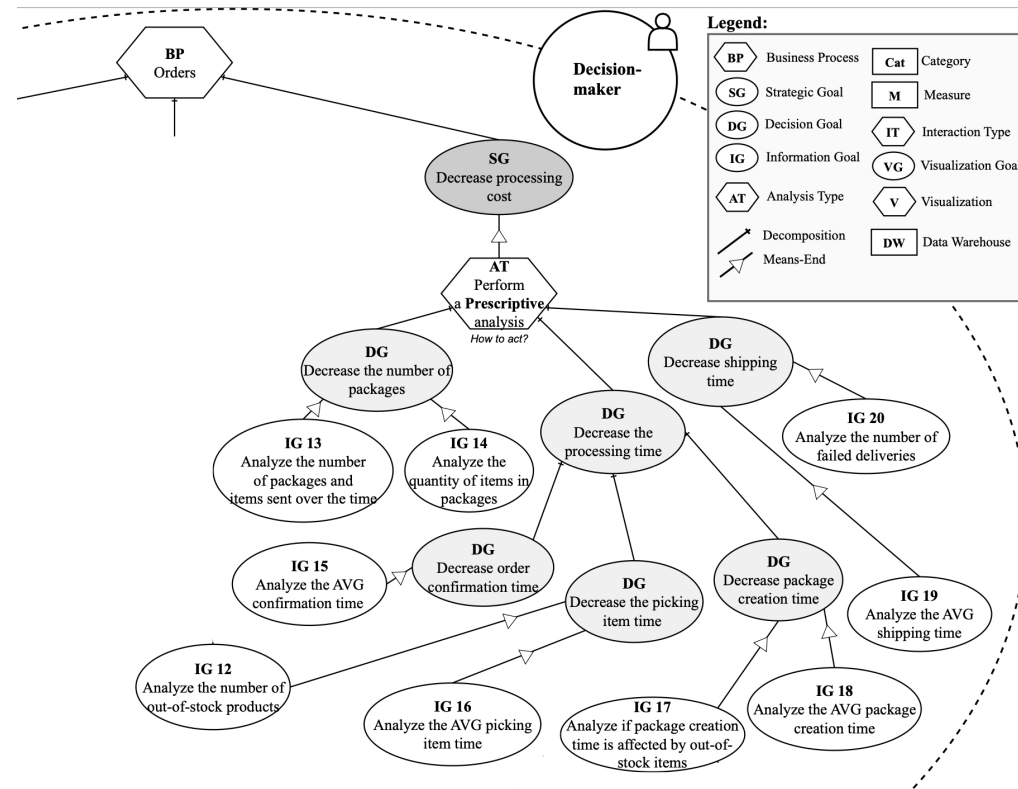
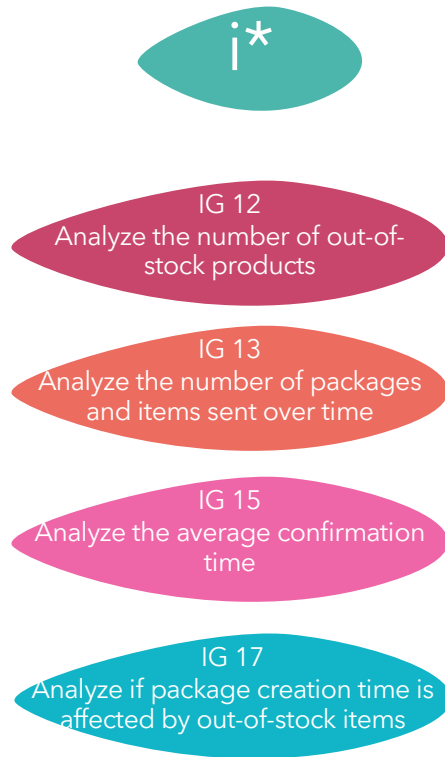


Strategic, Decision and Information Goals for a Data Warehouse system

- Strategic Goals (SGs) for business and operational processes
- SGs must be supported by Decision Goals (DGs) with business and operational concerns
- Information Goals (IGs) for the integrated analysis of business and operational indicators

A. Maté, J. Trujillo, X. Franch (2014). Adding semantic modules to improve goal-oriented analysis of data warehouses using i-star, Journal of Systems and Software 88. doi:10.1016/j.jss.2013.10.011.







Timestamp	Customer	Order	Product	Item	Package	Employee	Process (Manage ...)	Fact Table	Indicator	Aggregation Function
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IG 12
Analyze the number of out-of-stock products

IG 12	x		x	x	x			Activities	Out of Stock	Item Count	SUM/COUNT
-------	---	--	---	---	---	--	--	------------	--------------	------------	-----------

IG 13
Analyze the number of packages and items sent over time

IG 13	x							Packages	Packages	Package Count	SUM/COUNT
	x			x	x	x		Packages	Packages Detail	Item Count	SUM/COUNT

IG 15
Analyze the average confirmation time

IG 15	x		x					Orders	Orders	Days to Confirm [Confirm Order Date - Place Order Date]	AVG
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IG 17
Analyze if package creation time is affected by out-of-stock items

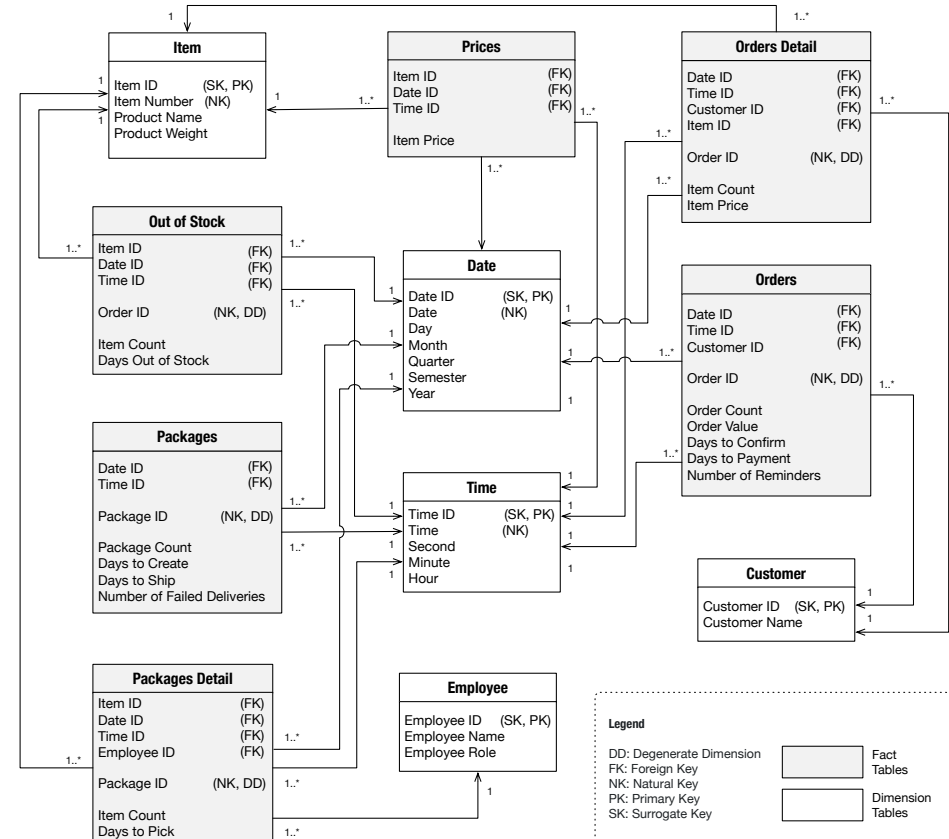
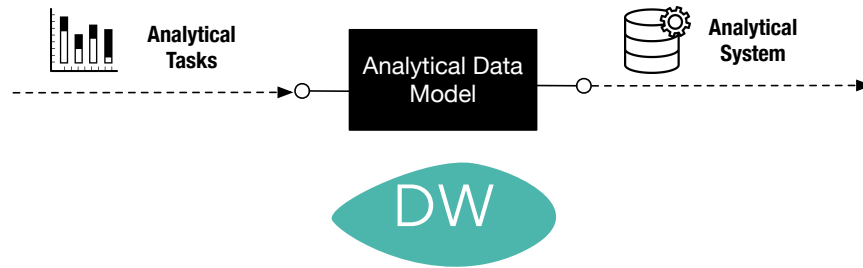
IG 17	x		x	x	x			Activities	Out of Stock	Days Out of Stock [Pick Item Date - Out of Stock Date]	AVG
-------	---	--	---	---	---	--	--	------------	--------------	---	-----

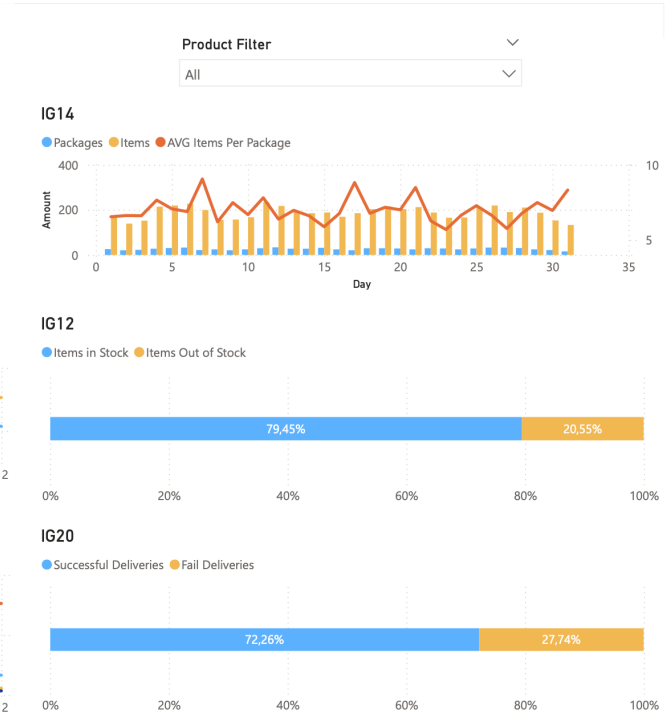
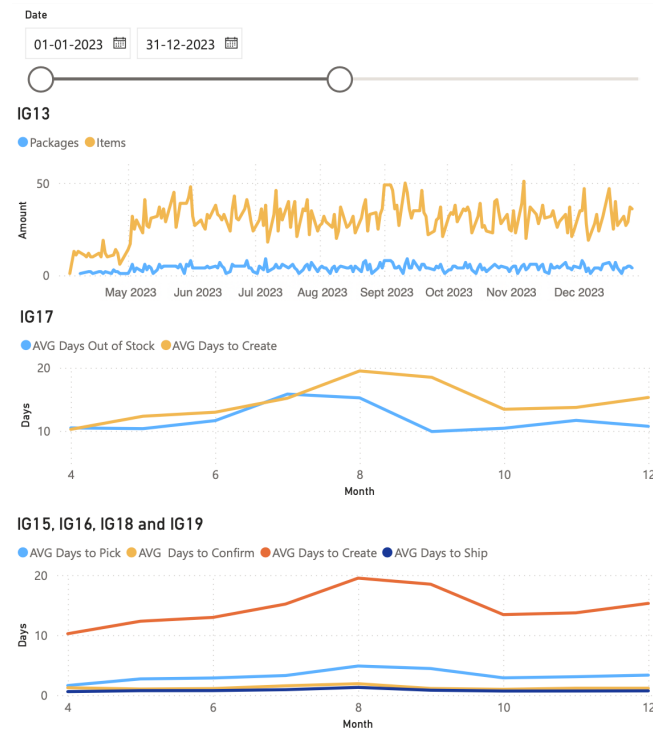
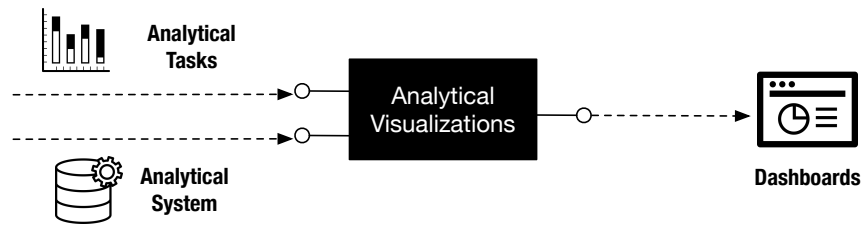


DW

Fact tables?
Level of
detail?

	Timestamp	Customer	Order	Product	Item	Package	Employee	Process (Manage ...)	Fact Table	Indicator	Aggregation Function
IG 1	x		x					Orders	Orders	Order Count	SUM/COUNT
IG 2	x		x					Orders	Orders	Order Value	SUM
IG 3	x	x	x					Orders	Orders	Days to Payment [Pay Order Date - Confirm Order Date]	AVG
IG 4	x	x	x					Orders	Orders	Number of Reminders	SUM
IG 5	x	x	x					Orders	Orders	Order Value	SUM
IG 6	x	x	x					Orders	Orders	Order Count	SUM/COUNT
IG 7	x	x	x	x	x			Orders	Orders Detail	Item Price	AVG
	x			x	x			Prices	Prices	Item Price	AVG
IG 8	x	x	x	x	x			Orders	Orders Detail	Item Count	SUM/COUNT
IG 9	x	x	x	x	x			Orders	Orders Detail	Item Count	SUM/COUNT
IG 10	x		x	x	x			Activities	Out of Stock	Item Count	SUM/COUNT
IG 11	x		x	x	x			Activities	Out of Stock	Item Count	SUM/COUNT
IG 12	x		x	x	x			Activities	Out of Stock	Item Count	SUM/COUNT
IG 13	x					x		Packages	Packages	Package Count	SUM/COUNT
	x			x	x	x		Packages	Packages Detail	Item Count	SUM/COUNT
IG 14	x			x	x	x		Packages	Packages Detail	Item Count	SUM/COUNT
IG 15	x		x					Orders	Orders	Days to Confirm [Confirm Order Date - Place Order Date]	AVG
	x			x	x	x	x	Packages	Packages Detail	Days to Pick [Pick Item Date - Confirm Order Date]	AVG
IG 17	x		x	x	x			Activities	Out of Stock	Days Out of Stock [Pick Item Date - Out of Stock Date]	AVG
	x					x		Packages	Packages	Days to Create [Create Package Date - (first) Pick Item Date]	AVG
IG 19	x					x		Packages	Packages	Days to Ship [Send Package Date - Create Package Date]	AVG
	x					x		Packages	Packages	Number of Failed Deliveries	SUM





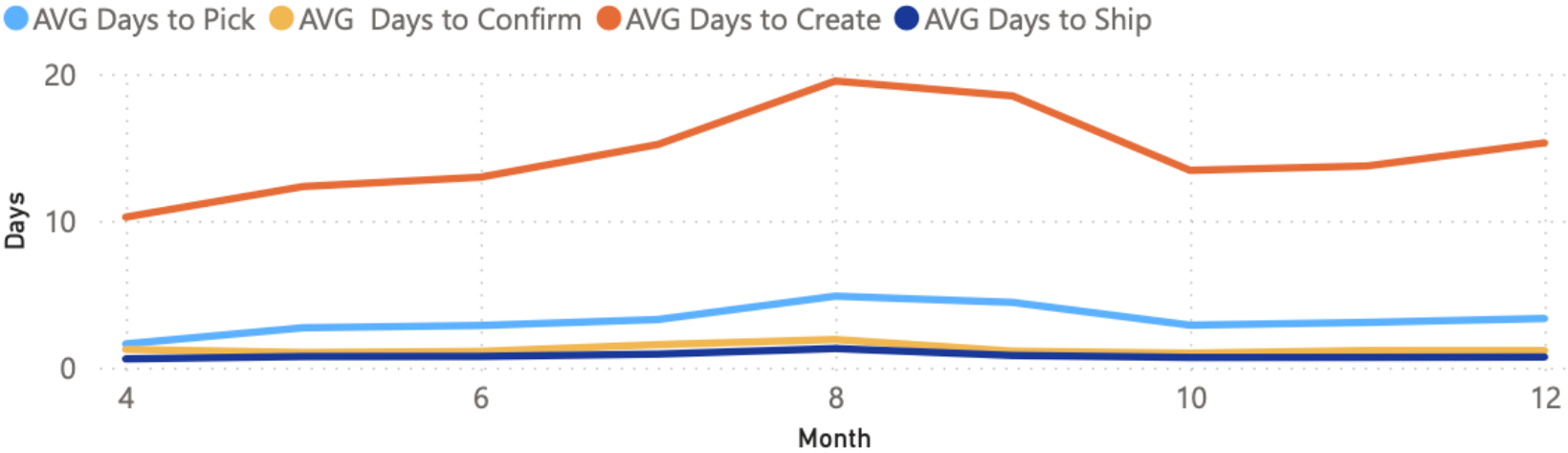
IG 15
Analyze the AVG
confirmation time

IG 16
Analyze the AVG picking
item time

IG 18
Analyze the AVG package
creation time

IG 19
Analyze the AVG shipping
time

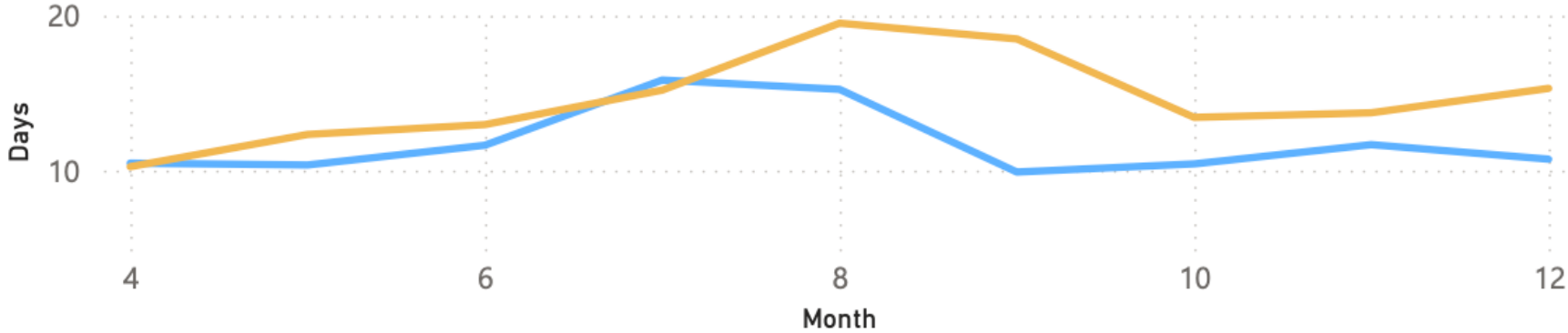
IG15, IG16, IG18 and IG19



IG 17
Analyze if package creation time is affected
by out-of-stock items

IG17

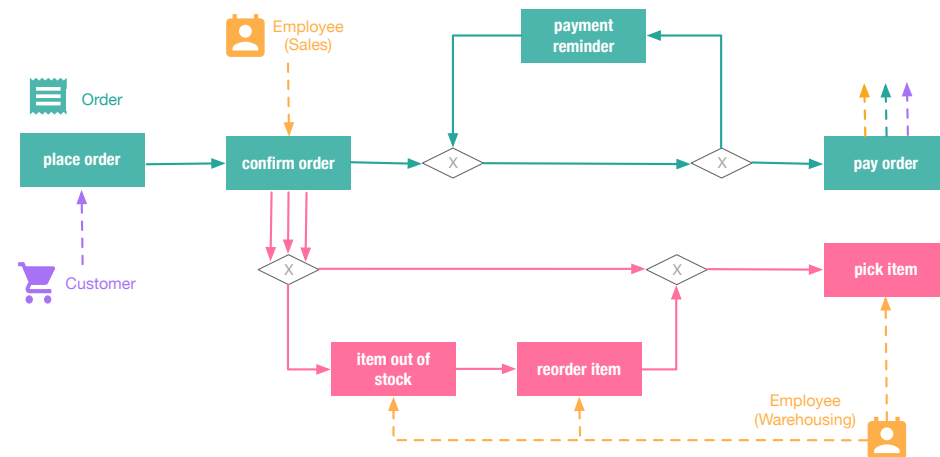
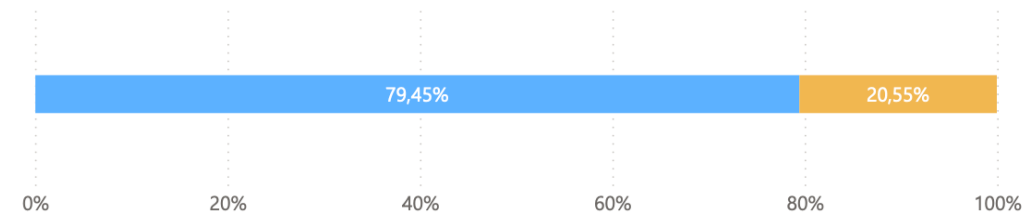
● AVG Days Out of Stock ● AVG Days to Create



IG 12
Analyze the number of out-of-stock
products

IG12

● Items in Stock ● Items Out of Stock

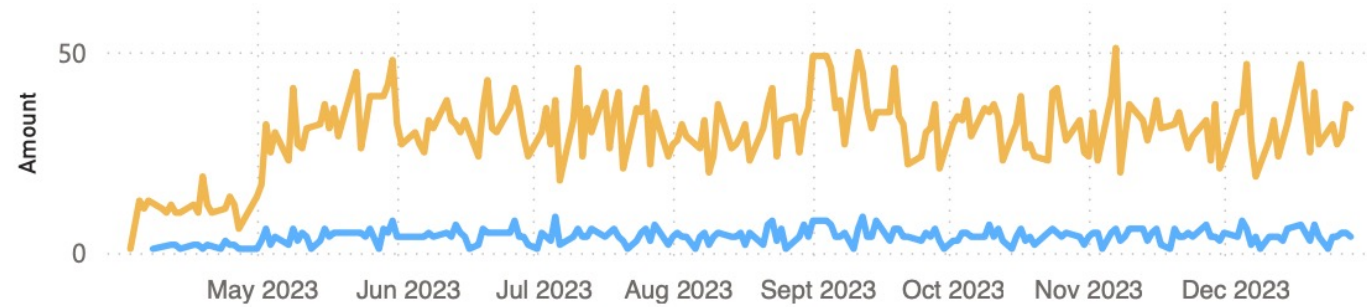


IG 13

Analyze the number packages and items sent over time

IG13

● Packages ● Items

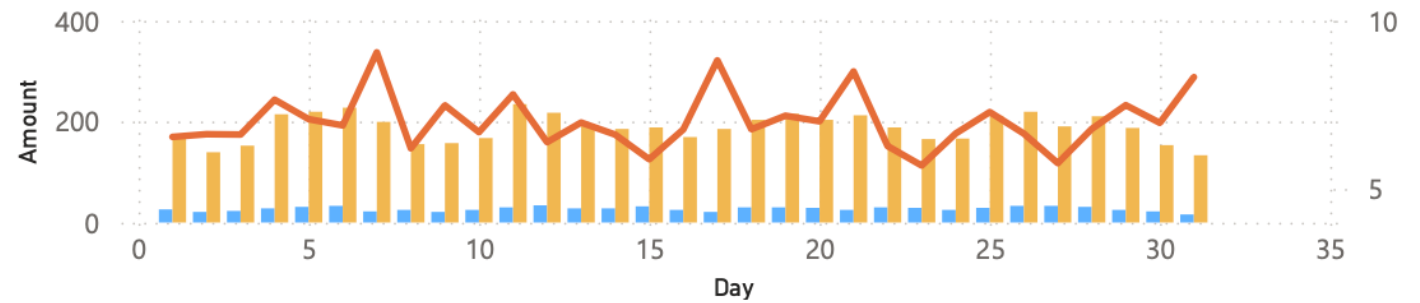


IG 14

Analyze the quantity of items in packages

IG14

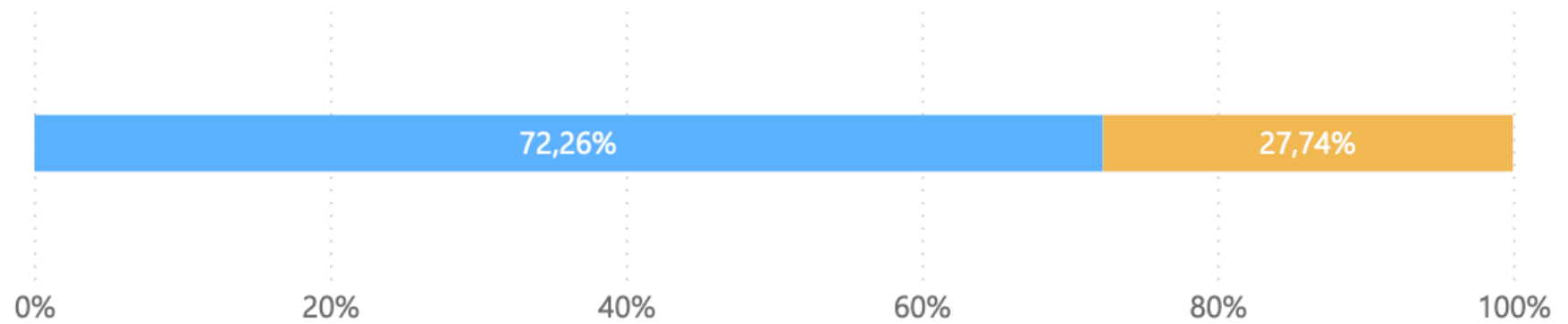
● Packages ● Items ● AVG Items Per Package



IG 20
Analyze the number of failed
deliveries

IG20

● Successful Deliveries ● Fail Deliveries



MORE INFORMATION

Integrating Business and Process Analytics for Enhanced Data-Driven Decision-Making

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Abstract

By integrating Business Analytics and Process Analytics, organizations can gain a deeper understanding of the relationship between process inefficiencies and business outcomes, leading to improved data-driven decision-making. This integration, however, is often overlooked, with limited methodological guidance for systematically combining these two analytical domains. Motivated by this challenge, this paper proposes a methodological approach for the identification of analytical visualizations that allow the aforementioned integration to be achieved systematically. The proposed approach is tested in a running example, which includes an instantiation of a Data Warehouse system for supporting data integration and analysis for business process analytics.

